

## PETISTRASK, Sweden

WMO: 022820

Lat: 64.5659N

Lon: 19.6943E

Elev: 846

StdP: 14.25

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -21.6      | -15.3 | -27.4                         | 1.2 | -21.5 | -20.2 | 1.9 | -15.2 | 12.4                  | 26.6 | 11.5 | 25.9 | 0.4                   | 270  | 0.620 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.6                   | 77.6            | 62.4 | 73.9 | 61.1 | 70.5 | 59.2 | 65.0                | 73.8 | 62.9 | 70.9 | 61.0 | 68.3 | 3.9                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 61.7                            | 84.7 | 68.3 | 59.6 | 78.7 | 66.2 | 57.8 | 73.5 | 64.1 | 30.4          | 74.0 | 28.8 | 71.0 | 27.5 | 68.7 | 72.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 12.0          | 10.0 | 8.9 | -27.5                      | 81.6 | 7.1                | 4.3 | -32.6                                              | 84.7 | -36.7      | 87.2 | -40.6      | 89.7 | -45.8      | 92.8 |    |    |
| (5)               |      |     | -28.1                      | 67.7 | 6.9                | 2.6 | -33.1                                              | 69.6 | -37.1      | 71.1 | -41.0      | 72.6 | -46.0      | 74.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 35.1   | 12.7  | 13.2  | 21.9  | 32.9  | 45.0  | 53.6  | 59.3  | 56.2  | 46.8  | 34.9  | 25.3  | 17.9  |  |  |
|                                                                        | DBStd       | 18.53  | 12.55 | 13.44 | 10.22 | 6.88  | 7.07  | 6.04  | 5.42  | 5.48  | 5.80  | 7.42  | 9.50  | 12.39 |  |  |
|                                                                        | HDD50       | 6133   | 1157  | 1029  | 870   | 514   | 187   | 30    | 1     | 10    | 128   | 469   | 742   | 996   |  |  |
|                                                                        | HDD65       | 10936  | 1622  | 1449  | 1335  | 964   | 619   | 346   | 192   | 278   | 545   | 933   | 1192  | 1461  |  |  |
|                                                                        | CDD50       | 696    | 0     | 0     | 0     | 0     | 33    | 137   | 291   | 202   | 33    | 0     | 0     | 0     |  |  |
|                                                                        | CDD65       | 24     | 0     | 0     | 0     | 0     | 0     | 3     | 16    | 5     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 306    | 0     | 0     | 0     | 0     | 7     | 65    | 184   | 50    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 40     | 0     | 0     | 0     | 0     | 0     | 8     | 30    | 2     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 3.6    | 3.1   | 3.4   | 4.3   | 4.1   | 4.5   | 4.0   | 3.3   | 3.0   | 3.2   | 3.7   | 3.4   | 3.3   |  |  |
|                                                                        | PrecAvg     | 23.5   | 1.4   | 1.3   | 1.1   | 1.2   | 1.5   | 2.4   | 3.9   | 3.0   | 2.2   | 2.0   | 1.9   | 1.6   |  |  |
|                                                                        | PrecMax     | 29.3   | 2.8   | 2.8   | 2.2   | 2.2   | 2.4   | 3.8   | 6.0   | 5.0   | 4.4   | 3.7   | 3.3   | 3.1   |  |  |
| (15) Precipitation                                                     | PrecMin     | 19.1   | 0.2   | 0.2   | 0.4   | 0.3   | 0.4   | 1.0   | 0.8   | 0.9   | 0.8   | 0.6   | 0.8   | 0.7   |  |  |
|                                                                        | PrecStd     | 3.5    | 0.6   | 0.6   | 0.5   | 0.6   | 0.7   | 0.8   | 1.4   | 1.3   | 1.3   | 0.9   | 0.8   | 0.6   |  |  |
|                                                                        | DB          | 38.5   | 41.3  | 46.3  | 60.7  | 74.6  | 80.6  | 83.9  | 79.0  | 66.6  | 53.9  | 46.4  | 39.5  |       |  |  |
|                                                                        | MCWB        | 36.4   | 37.6  | 39.1  | 47.5  | 56.9  | 62.8  | 66.6  | 65.1  | 57.7  | 49.3  | 44.4  | 37.0  |       |  |  |
|                                                                        | 2% DB       | 34.5   | 38.6  | 42.1  | 53.5  | 69.8  | 74.9  | 79.0  | 74.8  | 62.8  | 50.3  | 42.6  | 37.2  |       |  |  |
| (23) Mean Coincident Wet Bulb Temperatures                             | MCWB        | 33.3   | 35.3  | 36.8  | 43.4  | 54.8  | 59.6  | 64.3  | 62.7  | 55.5  | 47.5  | 41.3  | 35.7  |       |  |  |
|                                                                        | 5% DB       | 32.4   | 35.0  | 39.5  | 49.2  | 64.4  | 70.1  | 75.0  | 71.2  | 59.4  | 47.9  | 39.2  | 35.2  |       |  |  |
|                                                                        | MCWB        | 31.6   | 32.9  | 35.0  | 40.6  | 52.3  | 57.2  | 62.1  | 61.2  | 53.6  | 45.5  | 38.0  | 34.2  |       |  |  |
|                                                                        | 10% DB      | 30.1   | 32.0  | 36.7  | 45.4  | 59.4  | 66.4  | 71.6  | 67.1  | 56.6  | 45.7  | 37.0  | 33.3  |       |  |  |
|                                                                        | MCWB        | 29.5   | 30.7  | 33.2  | 38.3  | 48.8  | 55.3  | 60.7  | 59.3  | 52.3  | 43.8  | 36.1  | 32.6  |       |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 36.4   | 38.0  | 39.6  | 47.8  | 58.7  | 64.2  | 68.9  | 67.1  | 59.4  | 50.4  | 44.7  | 37.6  |       |  |  |
|                                                                        | MCDB        | 38.0   | 40.8  | 45.4  | 58.8  | 70.1  | 77.4  | 79.2  | 75.8  | 64.6  | 52.0  | 46.5  | 38.6  |       |  |  |
|                                                                        | 2% WB       | 33.5   | 35.6  | 37.4  | 44.4  | 56.0  | 61.0  | 65.9  | 64.4  | 56.7  | 47.8  | 41.4  | 35.8  |       |  |  |
|                                                                        | MCDB        | 34.1   | 38.3  | 41.5  | 52.7  | 68.0  | 71.6  | 74.8  | 71.8  | 60.7  | 49.7  | 42.4  | 36.8  |       |  |  |
|                                                                        | 5% WB       | 31.6   | 33.2  | 35.4  | 41.3  | 52.8  | 58.7  | 63.9  | 62.2  | 54.8  | 45.9  | 38.4  | 34.2  |       |  |  |
| (31) Mean Daily Temperature Range                                      | MCDB        | 32.2   | 34.7  | 38.5  | 47.8  | 63.1  | 67.4  | 72.5  | 68.7  | 58.1  | 47.6  | 39.1  | 34.9  |       |  |  |
|                                                                        | 10% WB      | 29.5   | 30.8  | 33.6  | 38.8  | 49.6  | 56.5  | 62.0  | 60.2  | 53.1  | 43.9  | 36.2  | 32.7  |       |  |  |
|                                                                        | MCDB        | 30.2   | 32.0  | 36.2  | 44.6  | 57.8  | 64.3  | 69.7  | 66.0  | 55.7  | 45.5  | 36.9  | 33.3  |       |  |  |
|                                                                        | 5% DB       | 15.8   | 18.3  | 19.6  | 18.4  | 18.3  | 17.5  | 17.6  | 16.6  | 15.1  | 10.8  | 9.9   | 14.1  |       |  |  |
|                                                                        | MCDBR       | 14.3   | 15.5  | 19.5  | 24.9  | 28.3  | 24.5  | 24.0  | 23.7  | 19.4  | 13.1  | 8.8   | 11.8  |       |  |  |
| (36) Clear-Sky Solar Irradiance                                        | MCWBR       | 13.7   | 13.6  | 15.0  | 15.4  | 14.8  | 11.6  | 11.0  | 12.3  | 12.6  | 10.2  | 7.9   | 11.3  |       |  |  |
|                                                                        | 5% WB       | 14.1   | 14.5  | 17.3  | 22.8  | 26.7  | 22.0  | 21.7  | 19.8  | 17.3  | 11.7  | 8.4   | 11.4  |       |  |  |
|                                                                        | MCWBR       | 13.7   | 13.1  | 13.6  | 14.6  | 14.6  | 11.1  | 10.7  | 11.0  | 11.8  | 9.7   | 7.8   | 11.0  |       |  |  |
|                                                                        | taub        | 0.194  | 0.266 | 0.296 | 0.329 | 0.335 | 0.337 | 0.349 | 0.337 | 0.307 | 0.290 | 0.337 | 0.351 |       |  |  |
|                                                                        | taud        | 1.972  | 2.187 | 2.252 | 2.346 | 2.473 | 2.494 | 2.473 | 2.501 | 2.568 | 2.434 | 2.695 | 2.728 |       |  |  |
| (42) All-Sky Solar Radiation                                           | Ebn at Noon | 114    | 203   | 250   | 268   | 278   | 279   | 272   | 265   | 252   | 200   | 108   | 55    |       |  |  |
|                                                                        | Edh at Noon | 8      | 18    | 26    | 30    | 30    | 30    | 30    | 26    | 20    | 14    | 7     | 4     |       |  |  |
|                                                                        | RadAvg      | 44     | 237   | 721   | 1253  | 1519  | 1679  | 1549  | 1176  | 698   | 279   | 67    | 15    |       |  |  |
| (45)                                                                   | RadStd      | 4      | 38    | 65    | 94    | 180   | 139   | 171   | 115   | 87    | 49    | 13    | 1     |       |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (44 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page